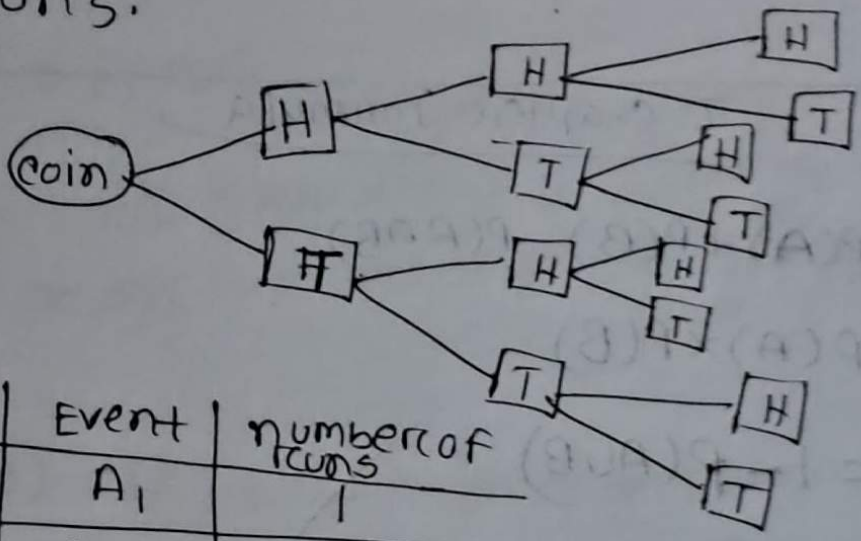
Exa - 7.26

95

For coin tossing ~~experiment~~ ~~Ex~~ ~~example 7.24~~, three times find the probability of Obtaining a) exactly two runs and b) less then two runs.

Sol:



OUTCOME	Event	number of runs
HHH	A_1	1
HHT	A_2	2
HTH	A_3	3
HTT	A_4	2
THH	A_5	2
THT	A_6	3
TTH	A_7	2
TTT	A_8	1

a) $P(X=2) = P(A_2)+P(A_4)+P(A_5)+P(A_7)$

$= 1/8 + \frac{1}{8} + \frac{1}{8} + \frac{1}{8} = \frac{4}{8} = 1/2$

b) $P(X<2) = P(A_1)+P(A_8)$

$= \frac{1}{8} + \frac{1}{8} = \frac{2}{8}$

$= 1/4$

Exa- 7.28] A leap year consists of 366 day with 29 days in February. If a leap year is selected at random, what is the probability that the selected leap year will consist of 53 saturdays?

Sol:

A leap year consists of 52 complete weeks (52 × 7) = 364 days. Plus two additional days with the following possible combination of the days of the week:

a) Sunday and monday
b) monday & tuesday
c) tuesday & wednesday
d) wednesday & thursday
e) thursday & friday
f) friday & saturday
g) saturday & sunday.

Now, two of the 7 combination above, saturday. Thus if 'A' is the event that one of these combination will consist of

saturday,
then

$$P(A) = 2/7.$$

Exa-7.35

Suppose a simple space consist of 500 person and are distributed according to their sex and employment status as shown in the accompanying table.

	employment status		
sex	Employed (E)	unEmployed (U)	Total
male (M)	255	20	275
Female (F)	80	145	225
total	335	165	500

Sol:

Given,

$n(S) = 500$

$n(M \cap E) = 255$

$n(F \cap E) = 80$

$n(M \cap E') = 20$

$n(F \cap E') = 145$

$n(F) = 225$

$n(M) = 275$

$n(E) = 335$

$n(E') = 165$

① A random chosen person will be male.

$$P(M) = \frac{n(M)}{n(S)}$$

$$= \frac{275}{500} = 0.55$$

② to get an unemployed person,

$$P(E) = \frac{n(E)}{n(S)}$$

$$= \frac{335}{500} = 0.67$$

③ The chosen one is male and employed as well :—

$$P(M \cap E) = \frac{n(M \cap E)}{n(S)}$$

$$= \frac{255}{500} = 0.51$$

④ male and unemployed as well :

$$P(M \cap E') = \frac{n(M \cap E')}{n(S)}$$

$= \frac{200}{500} = 0.04$

Now probability can be computed as a sum of the two joint probabilities:

$$P(M) = P(M \cap E) + P(M \cap U)$$
$$= 0.51 + 0.04$$
$$= 0.55$$

Ex – 7.36 In an office of 100 employees, 75 read English, 50 read Bangla dailies and 40 read both. An employee is selected at random. What is the probability that the selected employee.

a) Reads English newspaper?

b) Is at least one of the 11?

c) Reads none?

d) Reads Englis but not Bangla?

Sol:

There,

$n(E) = 75$

$n(B) = 50$

$n(E \cap B) = 40$

$n(S) = 100$

a) Read english newspaper,

$$P(E) = \frac{n(E)}{n(S)}$$

$$= \frac{75}{100} = 0.75$$

b) Read at least one of the papers,

$$P(E \cap B) = P(E) + P(B) - P(E \cap B)$$

$$= 75/100 + 50/100 - \frac{40}{100}$$

$$= 0.85$$

c) employee reads none,

$$P(\bar{E} \cap \bar{B}) = P(\overline{E \cup B})$$

$$= 1 - P(E \cup B)$$

$$= 1 - 0.85$$

$$= 0.15$$

d) The selected employee reads Bangla but not English is

$$P(B \cap E) = \frac{n(B) - n(B \cap E)}{n(S)}$$

$$= \frac{50 - 40}{100}$$

$$= 0.10$$

Ans

Exa - 7.38] The probability that a married man watched a certain TV show is 0.4 and that his wife watches the show is 0.5. The probability that a man watches the show, given that his wife does, is 0.7. Find,

a) the probability that a married couple watches the show.

b) The probability that wife watches the show given that her husband does.

c) The probability that at least one of the partners will watch the show.

Sol:

Given that,

$P(H) = 0.4$

$P(W) = 0.5$

$P(H|W) = 0.7$

a) The probability that the couple watches the show is:

$$P(W \cap H) = P(W)\, P(H|W)$$
$$= 0.5 \times 0.7$$
$$= 0.35$$

b) The probability that a wife watches the show given that her husband also watches

$$P(W|H) = \frac{P(W \cap H)}{P(H)} = \frac{0.35}{0.40}$$
$$= 0.875$$

c) The probability that at least one (either H or W or both) watches -

$$P(W \cup H) = P(W) + P(H) - P(W \cap H)$$
$$= 0.40 + 0.50 - 0.35$$
$$= 0.55$$

7.37 A pair of dice is thrown. Find the probability that sum of the points on the two dice is 10 or greater if a 5 appears on the first die.

Sol:

Let, A be the event that sum of the points on the two dice is 10 or greater and B be the event that a 5 appears on the first toss.

So, Probability $P(A|B)$.

Now,

$$A = \{(4,6),(5,5),(5,6),(6,4),(6,5),(6,6)\}$$

$$B = \{(5,1),(5,2),(5,3),(5,4),(5,5),(5,6)\}$$

$$A \cap B = \{(5,5),(5,6)\}$$

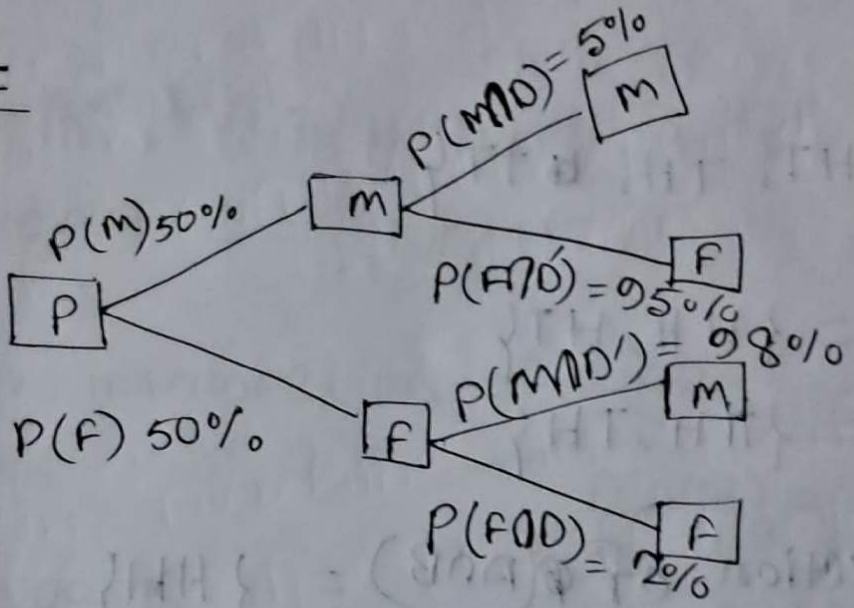
Hence

$$P(A|B) = \frac{2}{6}$$

$$= 1/3$$

Exa-7.43 In a community there are equal number of males & females. Suppose 5% of male and 2% of the females are disabled. A person is chosen at random. If this person is male, what is probability that he is disable?

Sol:-



Here,

$P(M) = 0.5$

$P(F) = 0.5$

$P(M \cap D) = 0.05$

$P(F \cap D) = 0.02$

Now, probability of the selected person is disable:

$$P(D|M) = \frac{P(D \cap M)}{P(M)} = \frac{0.05}{0.5}$$

$$= 0.1$$

Exa 7.44] Two ideal coins are tossed. Let A be the event 'head on the first coin' & B the event that 'head on the second coin'. A sample space for this experiment is, ⟶

Sol:

$S = \{HH, HT, TH, TT\}$

There,

A event $= \{HH, HT\}$

B event $= \{HH, TH\}$

Now intersection of $P(A \cap B) = \{HH\}$

$P(A \cap B) = \frac{1}{4}$

Now,

$P(A) = \frac{2}{4} = \frac{1}{2}$

$P(B) = \frac{2}{4} = \frac{1}{2}$

$\therefore P(A \cap B) = P(A) \times P(B)$

$= \frac{1}{2} \times \frac{1}{2}$

$\frac{1}{4} = \frac{1}{4}$

So, event A & B are independent.

A

Exa-7.45] Three coins are tossed. show that the events "heads on the first coin" and the event "tails on the last two" are independent.

Sol:

$S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$

There, Event $A = \{HHH, HHT, HTH, HTT\}$

event $B = \{HTT, TTT\}$

Now, intersection of $(A \cap B) = \{HTT\}$

$$P(A \cap B) = \frac{1}{8}$$

Now,

$$P(A) = \frac{4}{8} = \frac{1}{2}$$

$$P(B) = \frac{2}{8} = \frac{1}{4}$$

$$\therefore P(A \cap B) = P(A) \times P(B)$$

$$= \frac{1}{2} \times \frac{1}{4}$$

$$= \frac{1}{8}$$

$$\therefore = P(A \cap B)$$

So, event A and B independent.

Exa-7.46] A fire brigade has two fire engines operating independently. The probability that a specific fire engine is available when needed is 0.99.

a) what is the probability that an engine is available when needed?

b) what is the probability that neither is available when needed?

Sol:

There,

$$P(A) = 0.99$$

$$P(B) = 0.99$$

Now,

$$P(A \cap B) = P(A) \times P(B)$$

$$= 0.99 \times 0.99$$

$$= 0.9801$$

a) Here the event of $A \cup B$, so that,

$$P(A \cup B) = P(A) \times P(B) - P(A \cap B)$$

$$= 0.99 \times 0.99 - 0.9801$$

$$= 0.9999$$

b) This event is $\overline{A} \cap \overline{B}$, which equals $\overline{A \cup B}$, so that the probability is -

$$P(\overline{A \cup B}) = 1 - P(A \cup B)$$

$$= 1 - 0.9999$$

$$= 0.0001$$

Exa - 7.56] The probability that A will die within the next 20 years is 0.025, and that B will die within the next 20 years is 0.030. a) what is the probability that both A & B will die within the next 20 years? That A will die, but B will not die? That neither will die? That at least one will die?

Sol:

we assume that A and B are independ.

Now,

a) the probability that both A & B will die is,

$P(A \cap B) = P(A) \times P(B)$

$= 0.025 \times 0.030$

$= 0.00075$

A

b) The probability that A will die but B will not die is,-

$P(A \cap \bar{B}) = P(A) \times P(\bar{B})$

$= 0.025 \times 0.97$

$= 0.02425$

A

$P(\bar{B}) = 1 - P(B)$

$= 1 - 0.030$

$= 0.97$

c) The probability that neither A nor B will die is

$P(\bar{A} \cap \bar{B}) = P(\bar{A \cup B})$

$= 1 - P(A \cup B)$

$= 1 - \{P(A) + P(B) - P(A \cap B)\}$

$= 1 - \{0.025 + 0.030 - 0.00075\}$

$= 0.94575$ R.

c) The probability that neither A nor B will not qs —

$$P(\bar{A} \cap \bar{B}) = P(\bar{A}) \times P(\bar{B})$$
$$= 0.975 \times 0.970$$
$$= 0.94575$$

$P(\bar{A}) = 1 - P(A)$
$= 1 - 0.025$
$= 0.975$
$P(\bar{B}) = 0.970$

A

d) At least one of them will die

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$
$$= 0.025 + 0.030 - 0.00075$$
$$= 0.05425$$

A

Exa – 7.61 | It is reported that for the Bangladesh adult population as a whole, 50 percent are above normal weight, 25 percent have high blood pressure, and 65 percent either are above ideal weight or have high blood pressure. If A is the event that a randomly chosen person is above his/her, normal

weight and B is the event that this person has high blood pressure, are A & B independent?

Sol:

Here,

$P(A) = 0.50$

$P(B) = 0.25$

$P(A \cap B) = P(A) \times P(B)$

$= 0.50 \times 0.25$

$= 0.125$

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$= 0.50 + 0.25 - 0.125$

$= 0.65$

Now,

$\Rightarrow P(A) \times P(B)$

$\Rightarrow 0.50 \times 0.25$

$\Rightarrow 0.125$

and,

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$P(A \cap B) = P(A) + P(B) - P(A \cup B)$

$= 0.50 + 0.25 - 0.125$

$= 0.10$

$\therefore P(A) \times P(B) \neq P(A \cap B)$

Now, A & B are not independent.

Exa-7.68 In the submission of an income tax return, a tax-payer has applied for all allowance of certain expenses, for which the legitimacy is subject to verification. A three-member revie board operating on a rotation schedule receive application for consideration. The first boad, consisting mainly of members have vast experie approves only 25% of such claims without verification. The second board, consisting mainl of midterm individuals, approves 50% of such claims. The third board with relatively newer member dubious about refuting claims approve 70% of such cases. There is a 20% chance that the application will be considered by the first board, a 50% chance that it will be considered by the second board, and a 30% chance that it will be considered by the third board.

a) What is the chance that the application will be approved?

b) If the application is approved, what is the probability that it was considered by

i) the first board ii) the second board
iii) the third board?

Sol:

B_1 = first denote
B_2 = second denote
B_3 = Third denote

a) $P(A) = P(B_1 \cap A) + P(B_2 \cap A) + P(B_3 \cap A)$

$= P(B_1)P(A|B_1) + P(B_2)P(A|B_2) + P(B_3)P(A|B_3)$

$= 0.20 \times 0.25 + 0.50 \times 0.50 + 0.30 \times 0.70$

$= 0.51$

b) i) $P(B_1|A) = \frac{P(B_1)P(A|B_1)}{P(A)}$

$= \frac{0.20 \times 0.25}{0.51}$

$= 0.098$

ii) $P(B_2|A) = \frac{P(B_2)P(A|B_2)}{P(A)}$

$= \frac{0.50 \times 0.50}{0.51}$

$= 0.490$

iii) $P(B_3|A) = \frac{P(B_3)\,P(A|B_3)}{P(A)}$

$= \frac{0.30 \times 0.70}{0.51}$

$= 0.412$

Exa- 7.69] Three coins are tossed simultaneously set up a sample space for this experiment and obtain the probability that

a) All are heads

b) No heads occur

c) Two or more heads occur

Sol:-

The sample space is,

$S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$

a) Let, A event all are heads

$A = \{HHH\}$

$\therefore P(A) = \frac{n(A)}{n(S)}$

$= \frac{1}{8}$ Ans

b) Let, B be the event of no heads occurs,

$B = \{TTT\}$

$\therefore P(B) = \frac{n(B)}{n(S)}$

$= 1/8$ Ans

c) Let, C be the event of two or more heads occur,

$C = \{HHH, HHT, HTH, THH\}$

$\therefore P(C) = \frac{n(C)}{n(S)} = \frac{4}{8}$

$= 1/2$ Ans

Exa-7.70| Examination results of 150 students showed that 95 students passed mathematics 75 students passed economic and 135 students passed at least one of the above subjects. A student is selected at random. what is the probability that the student.

a) passed both mathematics and economic?
b) failed both the subjects?
c) passed mathematics but failed economics?

Sol: Here,

M = passed mathematics
E = passed economics
$\overline{M}$ = fail mathematics
$\overline{E}$ = fail economics

$n(M) = 95,$
$n(E) = 75$
$n(M \cup E) = 135$
$n(S) = 150$

Now,

$$P(M) = \frac{n(M)}{n(S)} = 95/150$$

$$= 19/30$$

$$P(E) = \frac{n(E)}{n(S)} = 75/150$$

$$= 1/2$$

The probability of passing at least one of the subjects is $P(M \cup E)$.

$$\therefore P(M \cup E) = \frac{n(M \cup E)}{n(S)}$$

$$= \frac{135}{150} = 9/10$$

a) $P(M \cap E) = P(M) + P(E) - P(M \cup E)$

$$= \frac{19}{30} + 1/2 - 9/10$$

$$= 7/30$$

b) $P(\overline{M} \cap \overline{E}) = 1 - P(M \cup E)$

$$= 1 - \frac{9}{10}$$

$$= 1/10$$

c) $P(M \cap \overline{E}) = P(M) - P(M \cap E)$

$$= 19/30 - 7/30$$

$= 2/5$

Exa-7.75 Two ordinary six-sided dice were tossed. set up a sample space for this experiment and hence find the probability that,

a) sum of the number on the two dice is 7.

b) Numbers on die 1 are greater than the number on die 2.

c) First die shows an even number.

d) Numbers on both the dice are the same. (i.e throwing a double).

e) Difference in the outcomes of two dice is 2.

Sol:

There,

$S = \{ (1,1), (1,2), (1,3), (1,4), (1,5), (1,6), (2,1), (2,2), (2,3), (2,4), (2,5), (2,6), (3,1), (3,2), (3,3), (3,4), (3,5), (3,6), \ldots$

(4,1), (4,2), (4,3), (4,4), (4,5), (4,6), (5,1),
(5,2), (5,3), (5,4), (5,5), (5,6), (6,1), (6,2), (6,3)
(6,4), (6,5), (6,6)}

~~Now, a) $P(x_1+x_2=7) = \{$~~

~~b) P(sum is 7) = $\frac{6}{36}$~~

a) $P(x_1+x_2=7) = \{(1,6), (2,5), (3,4), (4,3), (5,2), (6,1)\}$

There are 6 cases.

$\therefore P(\text{sum is } 7) = \frac{6}{36}$

$= 1/6$ ∴

b)

P(points on the 1 die are greater than the second) = $P(x_1 > x_2)$

There are $\Rightarrow \{(2,1), (3,1), (3,2), (4,1), (4,2), (4,3), (5,1), (5,2), (5,3), (5,4), (6,1), (6,2), (6,3), (6,4), (6,5)\}$

Now,

$P(x_1 > x_2) = \frac{15}{36} = 5/12$ ∴

c)

P(Even number),

There are => {(2,1), (2,2), (2,3), (2,4), (2,5), (2,6),
(4,1), (4,2), (4,3), (4,4), (4,5), (4,6),
(6,1), (6,2), (6,3), (6,4), (6,5), (6,6)
}

$$P(\text{Even number}) = \frac{18}{36}$$

$$= 1/2$$

d)

There are => {(1,1), (2,2), (3,3), (4,4), (5,5),
(6,6)}

Now,

$$P(x_1 = x_2) = \frac{6}{36}$$

$$= 1/6$$

e)

The event is $P(x_1 - x_2 = 2)$.

There are => {(1,3), (2,4), (3,1), (3,5),
(4,2), (4,6), (5,3), (6,4)}

$P(x_1 - x_2 = 2) = \frac{8}{36}$

$= 2/9$ Ans

Exa- 7.82 In a class of 180 students, all of whom took English and History, 15 failed history, 10 failed English and 5 failed both.

a) Find the probability that a student chosen at random failed history but passed English.

b) What is the conditional probability that a student, who passed English, also passed history?

Sol:

There,

$E = 170$ $\quad n(\bar{E} \cap \bar{H}) = 5$

$n(\bar{E}) = 10$

$H = 165$

$n(\bar{H}) = 15$

$n(S) = 180$

a) The probability that the student failed history but passed English.

$$P(\bar{H} \cap E) = \frac{n(\bar{H} \cap E)}{n(S)}$$

$$= \frac{10}{180}$$

$$= \frac{1}{18}$$

$n(\bar{H} \cap E) = n(\bar{H}) - n(\bar{H} \cap \bar{E})$

$= 15 - 5$

$= 10$

b) This is the case of conditional probability of the event H/E.

$$P(H/E) = \frac{n(H \cap E)/n(S)}{n(E)/n(S)} = \frac{n(H \cap E)}{n(E)}$$

$$= \frac{160}{170}$$

$$= \frac{16}{17}$$

Exa-7.86 A and B are two events associated with an experiment. Suppose that $P(A) = 0.4$, $P(A \cup B) = 0.7$. If $P(B) = p$, find the value of p for which i) A and B are mutually exclusive and ii) A and B are independent.

Sol:

i) For A and B to be mutually exclusive,

$$A \cap B = \emptyset$$

Here,

$$P(A \cup B) = P(A) + P(B)$$

$$0.7 = 0.4 + P$$

$$P = 0.3$$

$\therefore$ not equals.

ii) For A and B to be independent

$$P(A \cap B) = P(A) \times P(B).$$

Now,

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$0.7 = 0.4 + 0.3 - P(A \cap B)$

$P(A \cap B) = 0$

$P(A) \times P(B) = 0$

$P(A) \times P(B) = 0.4 \times 0.3$

$= 0.12$

$P(A \cup B) = 0.7$

$P(A) = 0.4$

$P(B) = 0.3$

$\therefore P(A \cup B) = P(A) + P(B) - P(A) \times P(B)$

$\Rightarrow 0.7 = 0.4 + P(B) - 0.4 \times P(B)$

$\Rightarrow 0.3 = P(B) - 0.4 \times P(B)$

$\Rightarrow 0.3 = P(B) \times 0.6$

$\therefore P(B) = 0.5$

A and B NOW,

$P(A \cap B) = P(A) \times P(B)$

$= 0.4 \times 0.5$

$= 0.2$

. . . . . .

conditional Probability

The Probability to occure event A given that even B has already occured.

$P(A|B) \rightarrow$ [B ঘেওয়া আগে A occure হবে]

[যেটা ঘেওয়া থাকবে আগে সেটা পাবে হবে]

$\Rightarrow P(A|B) = \frac{P(A \cap B)}{P(B)}$

$\Rightarrow P(A \cap B) = P(A|B) * P(B)$

~~$P(A|B)\ P(B)$~~

~~$P(A \cap B)$~~

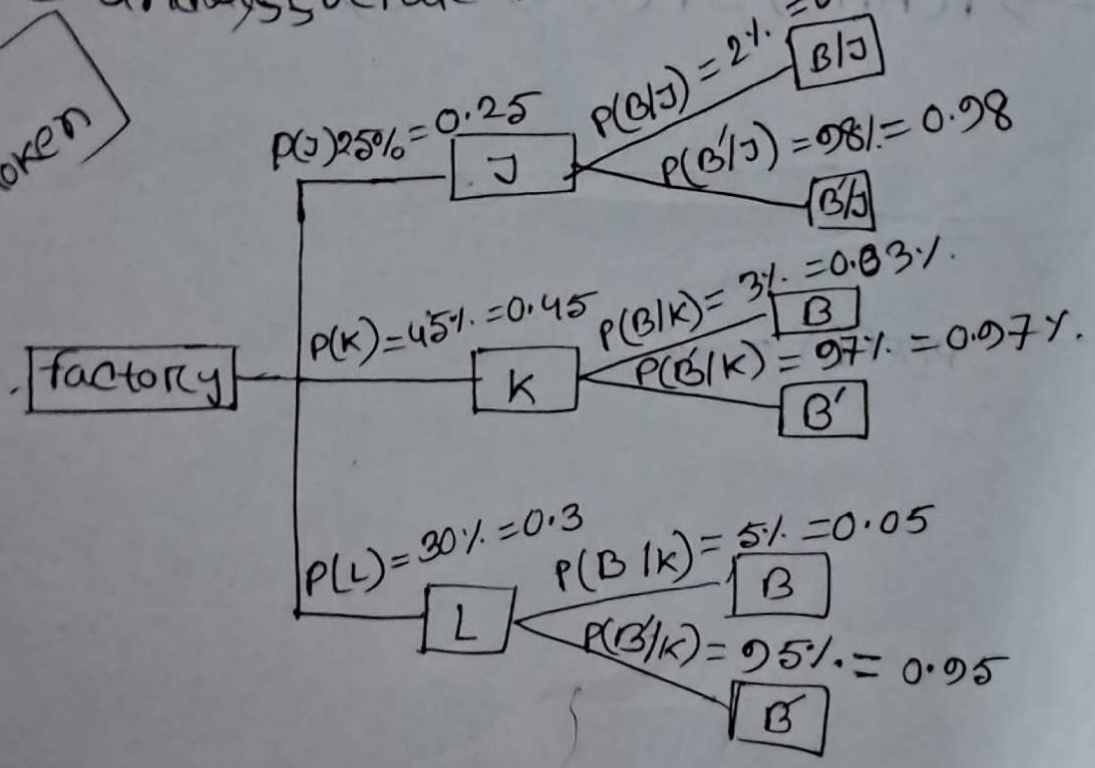
~~$P(A \cap B) = P($~~

Problem-1 In factory, three machines, J, K, L are used to make biscuits, machine J makes 25% of the biscuits. machine K makes 45% of the biscuits. The rest of the biscuits are made by machine L. It is known that 2% of the biscuits mad by machine J are broken, 3% of the biscuit made by K are broken and 5% the biscuits made by L are broken.

a) Draw a tree diagram to illustrate all possible outcome and associate probability.

sol:-

B = Broken
B' = not Broken



b) Calculate the probability that the biscuit is mad by machine J and is not broken.

$\Rightarrow P(J \cap B') = P(J) * P(B'/J)$

$= 0.25 * 0.98$

$= 0.245$

c) Calculate the probability that the biscuit is broken.

$P(B) \Rightarrow P(J \cap B) + P(K \cap B) + P(L \cap B)$

$\Rightarrow P(J) * P(B/J) + P(K) * P(B/K) + P(L) * P(B/L)$

$\Rightarrow 0.25 * 0.02 + 0.45 * 0.03 + 0.3 * 0.05$

$= 0.0335$

d) Given the biscuit is broken, find the probability that it was not made by machine K.

$\Rightarrow P(K'/B) = 1 - P(K/B)$

$= 1 - \frac{P(K \cap B)}{P(B)}$

$= 1 - \frac{0.45 * 0.03}{0.0335}$

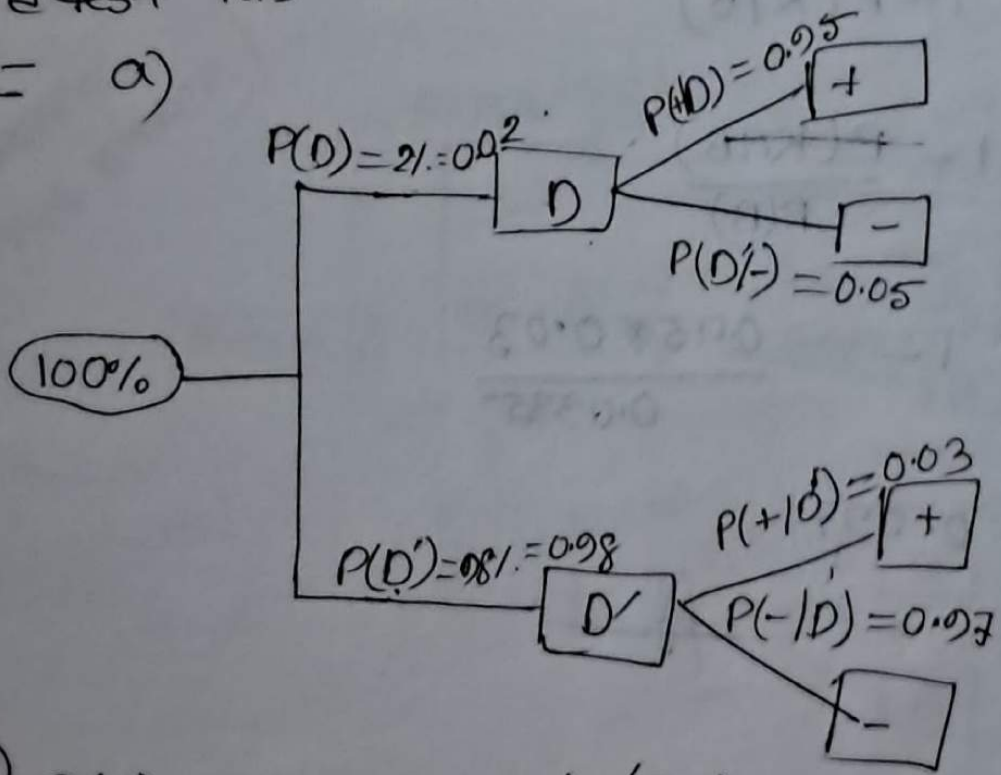
$= 0.597$

Pro-D| A disease is known to be present in 2% of a population. A test is developed to help determine whether or not someone has the disease. Given that a person has the disease, the test is positive with probability 0.95. Given that a person does not have the disease, the test is positive with probability 0.03

a) Draw a tree diagram to represent this information A person in selected at random from the population & test for this disease.

b) Find the probability that the test is positive. A doctor randomly selects a person from the population and test his for the disease. Given the test is positive

Sol:- a)



b) $P(+) = P(D \cap +) + P(D' \cap +)$

$= P(D) * P(D/+) + P(D') * P(D'/+)$

$= (0.02 * 0.95) + (0.98 * 0.03)$

$= 0.0484$

c) Find the probability that he does not have the disease.

c) $\Rightarrow P(D'/+) = \dfrac{P(D' \cap +)}{P(+)}$ (Given that person tested positive)

$= \dfrac{0.98 \times 0.03}{0.0484}$

$= 0.607$

e: Comment on the usefulness of this test.

d) $\Rightarrow P(+) = 0.0484$ and $P(D/+) = 0.607$

Not useful since $P(+)$ is very small when we would expect a high probability.

$P(D'/+)$ is high when we would expect a small probability.

Problem 3. The bag P contains 6 balls of which 3 are red and 3 are yellow. The bag Q contains 7 balls of which 4 are red and 3 are yellow.

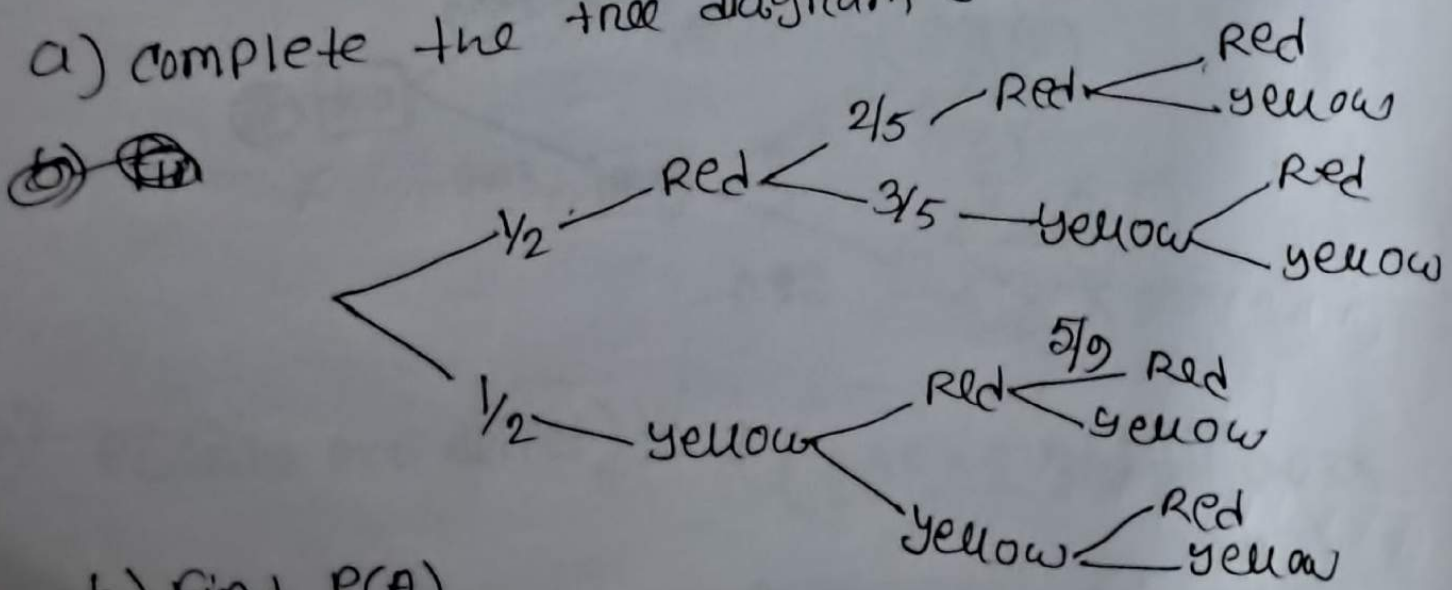
A ball is drawn at random from bag P and Placed in bag Q. A second ball is drawn at random from bag P and placed in bag Q.

A thirded ball is then drawn at random from the 9 balls in bag Q.

The event A occurs when 2 balls drown from bag P are of the same colour.

The event B occurs when the ball drawn from bag Q is red.

a) complete the tree diagram show below.



b) Find P(A)

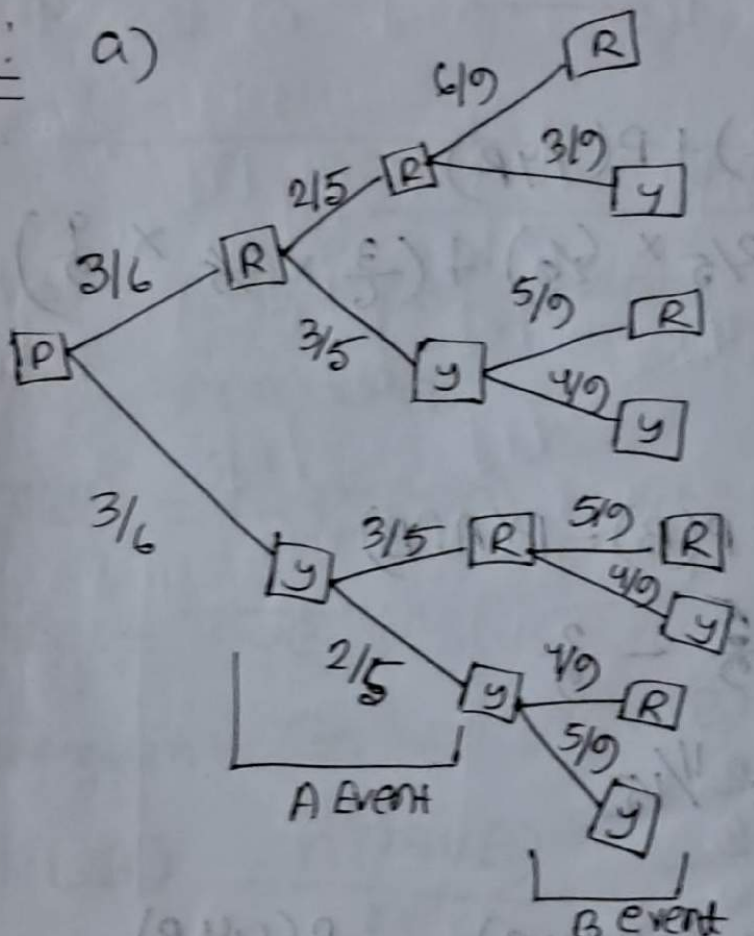
c) Show that P(B) = 5/9

d) Show that $P(A \cap B) = 2/9$

e) Hence find $P(A \cup B)$

f) Given that all three balls drawn are the same colour, find the probability that the all red.

Sol: a)



b) $P(A) = P(RR) + P(YY)$

$= \left(\frac{3}{6} \times \frac{2}{5}\right) + \left(\frac{3}{6} \times \frac{2}{5}\right)$

$= 0.4$

A

c) $P(B) = P(R,R,R) + P(R,Y,R) + P(Y,R,R) + P(Y,Y,R)$

$= \left(\frac{3}{6} \times \frac{2}{5} \times \frac{6}{9}\right) + \left(\frac{3}{6} \times \frac{3}{5} \times \frac{5}{9}\right) + \left(\frac{3}{6} \times \frac{3}{5} \times \frac{3}{9}\right) + \left(\frac{3}{6} \times \frac{2}{5} \times \frac{4}{9}\right)$

$= 5/9$

d) $P(A \cap B) = P(RRR) + P(YYR)$

$= \left(\frac{3}{6} + \frac{2}{5} \times \frac{6}{9}\right) + \left(\frac{3}{6} \times \frac{2}{5} \times \frac{4}{9}\right)$

$= 2/9$

e) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$= \frac{2}{5} + \frac{5}{9} - \frac{2}{9}$

$= 11/15$

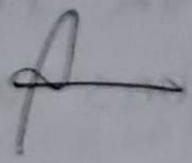
f) $P(\text{all } R \mid \text{all 3 same colour}) = \frac{P(\text{all } R)}{P(\text{all 3 same colour})}$

$= \frac{P(R,R,R)}{P(R,R,R) + P(Y,Y,Y)}$

$= \frac{1/2 \times (2/5) \times (2/3)}{1/2 \times (2/5) \times (2/3) + (1/2 \times 2/5 \times 1/5)}$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$= \frac{6}{11}$$



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7 Chapter Formula

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A \cap B) = P(A) \times P(B)$$

$$P(\overline{A \cup B}) = 1 - P(A \cup B)$$

$$\overline{A} \cap \overline{B} = \overline{A \cup B}$$

Independent হলে : $P(A \cap B) = P(A) \times P(B)$

$$P(A \cup B) = \frac{n(A \cup B)}{n(S)}$$

$$P(\overline{A} \cap B) = \frac{n(\overline{A} \cap B)}{n(S)}$$

$$n(\overline{A} \cap B) = n(A) - n(\overline{A} \cap \overline{B})$$